

Using Claude Code without human interaction: scripts, CI, pipes

Base flag: `-p`

The `-p` flag (or `--print`) activates non-interactive mode. Claude returns the response in the terminal and exits immediately without waiting for user input.

```
# Direct prompt
claude -p "Analyze this file and list the bugs"

# With pipe stdin (content becomes the input)
git diff | claude -p "Explain these changes"
cat error.log | claude -p "Identify the root cause"
```

Output formats

`--output-format` controls the response structure, which is essential for integration into scripts.

Format	Usage
text	Human-readable, default
json	Parseable by <code>jq</code>
stream-json	Real-time streaming

```
# JSON output for automatic parsing
git status --short | claude -p "Categorize the changes" \
  --output-format json | jq -r '.result'

# Stream JSON for long operations
cat report.txt | claude -p "Summarize" --output-format stream-json --verbose
```

Flow control in pipes

In print mode, `claude -p` waits for the complete response and writes it to stdout in a single block, which simplifies pipes with tools that expect complete input. Real-time streaming is opt-in via `--output-format stream-json`.

```
claude -p "Generate a report" > report.md
```

CI/CD usage

In an isolated container, `--dangerously-skip-permissions` suppresses all confirmation requests. Reserve exclusively for sandboxed environments where Claude cannot access sensitive resources.

```
# GitHub Actions (isolated container)
claude -p "Run the tests and fix failures" \
  --dangerously-skip-permissions \
  --output-format json
```

Common patterns

```
# Automated log analysis
tail -n 200 app.log | claude -p "Critical alerts only" \
  --output-format json

# PR review (package.json integration)
git diff main...HEAD | claude -p "Security review, JSON format" \
  --output-format json > review.json
```

Interactive vs non-interactive

Aspect	Interactive	Non-Interactive (<code>-p</code>)
Output	Stream UI	Raw stdout
Permissions	Prompts or saved rules	Configure allow rules or a bounded permission mode
Usage	Development	CI, scripts, pipes
Session	Persistent	Saved by default; can continue or resume

Best practice: limit pipe size to avoid exceeding the context window. Filter with `head`, `grep` or `--name-only` before sending to Claude.